

2. After maxing out your tax-exempt account, optimal allocation decisions can be quite complicated. They can depend on time horizon and tax rates. Different investment vehicles will have different tax implications. The tax treatment interacts with the returns the assets generate to make the joint optimal allocation and location decision complex.
3. Individuals need to focus on after-tax returns, not pre-tax returns. Unfortunately, most of the financial industry only reports pre-tax returns.

The first lesson is crucial. You can save directly in a tax-deferred pension. Or you can find an asset that pays you a tax-exempt return. One extremely important asset class that does this is municipal bonds, which allow money to accumulate tax-free in a taxable account.

3. Municipal Bonds

3.1. CHARACTERISTICS OF THE MUNI MARKET

Municipal bonds, or munis, are bonds issued by states, municipalities, 501(c)(3) nonprofit corporations (named for the section of the Internal Revenue Code granting them a tax exemption), and other tax-exempt entities. This market is large: at the end of 2011 there were \$3.7 trillion in muni securities outstanding, about a third the size of the U.S. Treasury market. The coupons and original issue discount (OID) on regular munis are exempt from Federal income tax. In addition, individuals are exempt from state income tax on muni coupons and OID on bonds issued by tax-exempt authorities in the investor's home state. But some munis are subject to federal income tax and the alternative minimum tax, so individuals have to be careful.⁶

The tax-exempt muni market is unique to the United States. In other countries, investors holding state bonds pay national income taxes. Munis are exempt from federal taxes here due to the separation of the federal and state governments in the constitution (federalism). In *Pollack v. Farmers' Loan and Trust* (1895), the Supreme Court interpreted the constitution as not allowing the Federal government to tax the states. Put simply, one government could not tax another. But in 1988 the Supreme Court overturned the constitutional basis for the tax exemption in *South Carolina v. Baker*, so the muni exemption is now the purview of Congress. Periodically there are calls to rescind it, but it will take an extremely brave Congress to do so.⁷

⁶ There are taxable munis, of which one important class is Build America Bonds. See Ang, Bhansali, and Xing (2010a).

⁷ The calls to rescind the tax exemption of municipal debt are not a recent phenomenon: Ott and Meltzer (1963) report that there were 114 resolutions to repeal the municipal tax exemption from 1920 to 1943.